

# Exploratory Analysis of Audio Features for Dominant Feature Identification: A Multimodal Machine Learning Evaluation of Song Popularity Prediction Using Deep Neural Networks, Clustering, and Regression Models

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## Abstract

With the vast collections of artist catalogues, music streaming services have become more popular and accessible, and technology has helped to shift the focus away from physical purchases. However, picking which songs will go viral is still a tough job since audiences change all the time. The study analyzes song popularity prediction within the context of Original Pilipino Music (OPM) through various machine learning models, specifically focusing on dominant audio features that influence listener engagement. Key findings reveal that attributes such as tempo, energy, loudness, and danceability strongly correlate with popularity, with high-energy and rhythmically dynamic features often associated with popular tracks. The research involved clustering techniques, feature extraction, and dimensionality reduction (such as UMAP and LLE) to identify patterns, demonstrating that mainstream and niche song clusters could be distinguished by their distinct audio profiles. The Deep Neural Network (DNN) model, enhanced with an attention mechanism, showed improved prediction capabilities for song popularity, leveraging these dominant features effectively. In terms of model performance, the DNN achieved high accuracy, underscoring the potential of machine learning in capturing the audio elements that appeal broadly across OPM listeners.

## CCS Concepts

• Information systems; • Information retrieval; • Evaluation of retrieval results; • Presentation of retrieval results;

## Keywords

Audio Features, Deep Neural Networks, Attention Mechanism, Clustering, Feature Extraction, Machine Learning

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## 1 Introduction

The global music industry has undergone a seismic shift driven by streaming platforms which now hold most of music revenue. With their immense libraries of artist catalogues, technology has worked to move the burden away from physical purchases. However, picking which songs will go viral is still a tough job since audiences are not static. Each year, technology brings the music industry closer to AI Integration. This phenomenon has now touched every aspect of music from when it is created to when it is distributed. This has great significance for audio features (that provide appeals to listeners), especially unique feature(s) in regional markets, profit from AI, powerful commercials [1], etc.; thus, a critical understanding of such switching landscapes is imperative. Musical taste is similar in that it is conditioned often by the environment, the context, and what an individual has gone through in life. Music prediction algorithms and their validity are difficult to assess, as the matter of good or bad preferences is very subjective [2]. The chorus, specifically, is one structural feature of popular music that is particularly important, since previous work finds it to be the “most catchy” and “most salient” section of a song. While there are those that go beyond merely assessing the audio of pop songs, this study limits its scope to the choruses of the songs and attempts to ascertain how these elements contribute to listener appeal.



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Previous works on predicting the popularity of songs focus on general classification models that classify songs as popular or not popular over global datasets [3]. Yet, no study conducted with a purposeful investigation of relevant audio features and their contribution towards popularity in specific regional contexts, like the Philippine market. There is also an opportunity to use a DNN as a feature impact analysis prototype instead of using it as an overall popularity predictor. With this approach, we can utilize the DNN model in a new way—through an attention mechanism to indicate the influence of features—to explain the feature interactions, as well as the relative importance of features contributing to song popularity.

Our approach bridges the gap between technical innovation and commercial application. For an industry increasingly driven by AI, this framework offers a scalable solution to mitigate financial risks and a competitive advantage in the Philippines' ever-evolving music ecosystem.

## 2 Review of Related Literature

### 2.1 Audio Feature-Based Classification

In a study [4], an automatic OPM song genre classification system is reported with the goal of improving hit songs for specific audience targeting. Targeting ballad, alternative rock, and rock as three genres, the researchers collected 380 OPM MIDI files and extracted 56 musical features. The classifiers are Linear Discriminant Analysis, Classification and Regression Trees (CART), and Neural Networks. It looks at each method in some detail (e.g., the discriminant functions used by LDA; the tree and impurity reduction in CART; backpropagation through gradient descent, etc.). With a train/test split of 80/20, the study was able to reach an accuracy of close to 100% classification with LDA showing the best results. They also included confusion matrices along with success rates for each classifier, demonstrating further potential of LDA and NN for accurate cross-genre classification, particularly with respect to melodic intervals. This indicates the potential application of both application of genre-based playlist creation and audience targeting in the music industry.

### 2.2 Use of Machine Learning in Predicting Popularity

For song hit prediction, in [6] audio features and sentimental lyrics were analyzed by applying machine learning models in R Studio. Billboard Hot 100 Charts Spotify Open Lyrics Datasets This consisted of extracting and pre-processing the data, followed by feature selection, to create a prediction model. Confusion matrices were used to evaluate 5 different machine learning algorithms on C4, Naïve Bayes, and Logistic Regression output accuracies Additionally, the weights of the parameters from Logistic Regression were additionally analyzed to determine its effect on the model performance. One of the best features for predicting hits was danceability. In a wider discussion of its results compared to previous studies (some of which had the exact same results as the new study), the study also cautioned that although the results would be helpful in a sound and lyrical feature-based 'hit-song formula', to ever come up with a final solution is not straightforward. One of the findings of

the study is consistent with its researcher's intent to examine dominant features is the important role of danceability for the models to predict popularity. There are more extensive prediction models, of course, but individual song features will have an iconic effect on hit songs.

The work in [7] also approached song popularity prediction within streaming services with a focus on the Top 50 Global Chart of Spotify and framed it as a classification problem by predicting whether a song will enter the chart or not, given the time frame of a certain interval. The authors categorized predictive methods into three generic types, social network data, acoustic data, and historical chart data. We gathered daily acoustic features (energy, tempo, key, etc.) for every song that made the Top 50 at least once between November 2018 and April 2019 through Spotify's Web API. Multiple rounds of predictions were made using ensemble methods (AdaBoost and Random Forest), probabilistic methods (Bernoulli and Gaussian Naïve Bayes), and various SVM kernels. Accuracy score of 89.09% when predicting a song's position in days ahead of time [5]. This shows why the research is relevant for artists, labels & the music business, how the methodology can be applied to different data types, and how it might improve in the future.

### 2.3 Deep Learning Techniques for Audio Categorization

The paper in [8] studies how AI and machine learning can be used to determine song popularity based on features such as genre, chords, and lyrics sentiment analysis. This work is a deviation from the conventional approaches of evaluating the popularity of music based on daily media platforms towards audio features and lyrics sentiments. Based on artist familiarity data using two datasets (one dataset consisted of Spotify song metadata and lyrics, whilst the other consisted of artist familiarity values), the study conducted sentiment and 14 profanity analyses to improve the efficiency of the dataset. We compared six algorithms and conducted hyperparameter tuning for optimum accuracy. The Spotify API was used to extract metadata while the Lyrics Genius API was used for web scraping, allowing real-time predictions to be done. Cross-validation with k-fold and visualizations were used to explore the effects of the input variables on track popularity. Using the Random Forest Classifier, which returns an accuracy of 85.06%, the study has identified tempo, loudness, acoustics, and liveliness as top features affecting song popularity. The use of sentiment analysis instead of profanity was also highlighted, and genre, key, and mode were of less importance in this model. They noted limitations and areas for further research, such as songs in languages other than English and more artist features to enhance predictability. The dataset was filtered down to only contain songs from the genres that we focus on which included 200,000 records and 52 genre labels. The models that were utilized in the study include Decision Trees, Naïve Bayes classifiers, linear SVMs, and Deep Neural Networks (DNNs), RNNs results showed that the Naïve Bayes classifier was the fastest classifier to train, with only 2.18 seconds for the balanced set. The performance from the neural network and RNN are comparably close together, but again, the RNN did not help boost the accuracy of the original NN. Neural networks showed the best overall performance while

decision trees performed the worst, Meanwhile, SVM and Naive Bayes classifiers gained higher Average Precision values.

### 3 Methodology

This section will discuss the methodology used and created to isolate and determine the audio features with the most influence over a song gaining popularity in the Philippines. The methodology consists of four major components: Data Collection and Preprocessing, Feature Extraction, Multi-Model Analysis, and Model Deployment and Evaluation. Each of these components offers a systematic methodology to dissect the key components that make music appealing to listeners. Every phase has been designed to ultimately serve the study's objective, which is to investigate the trends of the feature importance in popular music trends, rather than building an all-encompassing predictive model. This approach offers a culturally relevant and data-driven framework for understanding regional music trends by isolating and analyzing dominant characteristics in songs that have already been shown to be popular.

#### 3.1 Dataset Collection

The data collection process begins by acquiring a meticulously curated dataset from a professional data scientist, which comprises data about Spotify's top trending songs from 2017 to 2023. This dataset focuses on the top 200 songs each year over a span of five years (2018-2023). The present dataset for 2024 was derived from Spotify's weekly top 200 charts, serving to supplement the historical dataset, as it reflects the latest in song popularity. The dataset has also explicitly the top 200 songs from the week of '2024-10-28' so the analysis can reflect the recent changes in listeners' interests. This five-year period is significant, as previously mentioned [9], it is a crucial time in the development of the history of music streaming and consumption in the Philippines. The time frame was chosen to observe trends in Music 29 popularity during the pre-pandemic, pandemic, and post-pandemic eras. This also comes at a time when there has been a lot of progress in the local OPM scene, and streaming platforms in the country itself maturing as well. Preprocessing — one of the most crucial stages that includes several data operations like cleaning, normalization, and data transformation to ensure consistency and reliability. Data cleaning also eliminates inconsistencies, duplicity, and missing values giving you a more precise dataset. Normalization techniques are also done so it can therefore be possible to standardize all the values and compare data points to one another. This step helps prepare the dataset for further analysis and eliminates all the noise or irrelevant data to help facilitate an easier representation of the trends in popular music.

#### 3.2 Data Preprocessing

Data preprocessing involved cleaning, normalizing, and structuring the dataset for analysis, with additional integration of Spotify and YouTube API data to obtain audio features and select social metrics. In this phase, raw MP3 files were processed to focus specifically on the chorus segments by calculating a time similarity matrix, ensuring that analysis centers on the most memorable song components.

**3.2.1 API Integration Layer.** The API Integration Layer represents a crucial step, as it will allow for more features to come, including

audio and social engagement data. This layer is built on top of Spotify and YouTube APIs to fetch some useful high dimensional data for every song. Audio features: The audio features extracted from the Spotify API are listed below; The tempo, energy, danceability, and other characteristics about the song's sound profile. Also, similar to Twitter but with more granular social media engagement metrics per track, the YouTube API provides access to the YouTube view counts, determining the number of times a tune has been viewed, which also means how popular and how far a certain tune has gotten through listeners [10].

**3.2.2 Audio Processing.** Once the data has been obtained, audio processing steps become the main attention. In this step, we download a raw audio file for each of the songs in the 1,000 gathered in the dataset, i.e., download mp3 files. These files are the building blocks for more in-depth feature analyses and chorus extraction stages. The audio processing phase converts each file so that they can be analyzed consistently in terms of format, sample rate, and file corruption. This may also include normalization of levels as well as deletion of metadata which can interfere with extraction of features. Quality audio data provision is crucial for this analysis at ensuring contained tracks are representative of 31 their distinctive characteristics which is vital for reliable feature extraction and modeling in later stages.

**3.2.3 Chorus Extraction.** The chorus detection algorithm in this study aims to isolate the most repeated section of each song, typically the chorus, which is often the most emotionally resonant and memorable part. The process begins by transforming the audio into a chromagram, which represents pitch over time and simplifies pattern recognition. This chromatic representation is then used to construct a time-time similarity matrix, identifying segments of the song that are similar to each other. To enhance detection precision, the algorithm also employs a time-lag similarity matrix, optimizing the recognition of repeated patterns. After constructing these matrices, thresholding and denoising are applied to eliminate minor repetitions and highlight significant, repetitive sequences. Finally, the algorithm identifies and counts these repeated segments, selecting the most frequent and strongest similarity scores as the best candidate for the chorus, thereby ensuring a focused analysis on the song's most impactful portion.

#### 3.3 Feature Extraction

As shown in Table 1, the researchers extracted 16 main audio features after processing the audio, which we used for the next analysis. Features such as danceability, energy, loudness, etc., each contribute uniquely to understanding the appeal of a song. The feature extraction focuses on sounds that correlate with the trend in pop music. The Librosa library was employed to load each song, which subsequently calculates key audio features listed in the table below. With every feature extracted, they are stored in the Data Frame with each row for each song.

The preprocessing steps ensure the data's suitability for machine learning analysis. Interquartile Range (IQR) filtering eliminates outliers by removing values beyond 1.5 times the IQR from the first and third quartiles, ensuring a more homogeneous dataset. Min-Max Scaling standardizes features to a range of 0 to 1, maintaining

**Table 1: Key Audio Features Extracted**

| Audio Feature        | Value                                                                          |
|----------------------|--------------------------------------------------------------------------------|
| Key                  | Indicates the song’s harmonic scale, affecting its mood and structure          |
| melody_pitch_profile | Captures the contour and emotional quality of the melody                       |
| tempo                | Measures the song’s speed in beats per minute, impacting energy                |
| beat_strength        | Reflects the intensity of the beat, influencing rhythm                         |
| chord_progressions   | Represents the sequence of chords, shaping harmony                             |
| timbre               | The unique tone quality, distinguishing different sounds                       |
| loudness             | Measures the overall volume, indicating intensity                              |
| energy               | Shows the intensity and activity level, with higher values for livelier tracks |
| spectral_centroid    | Indicates sound brightness, with higher values for sharper tones               |
| spectral_bandwidth   | Measures frequency range, with wider bandwidth for richer sound                |
| zero_crossing_rate   | Reflects signal noisiness, with higher rates for percussive sounds             |
| danceability         | Shows how suitable a track is for dancing                                      |
| instrumentalness     | Shows whether a track contains no vocals                                       |
| valence              | Measures the musical positiveness of a track                                   |
| acousticness         | Measures whether a track is acoustic or not                                    |
| liveness             | Shows whether the track was performed live or not                              |
| Audio Feature        | Value                                                                          |

consistency across attributes for comparability. Categorization organizes YouTube view counts into four classes based on thresholds, facilitating structured analysis of popularity levels. Oversampling, utilizing the SMOTE algorithm, balances the dataset by increasing underrepresented categories, ensuring robust training and reliable model outcomes.

### 3.4 Model Analysis

This phase centers on feature importance, clustering, and dimensionality reduction techniques to pinpoint dominant attributes for song popularity. Correlation analysis identifies relationships among features, while mutual information reveals both linear and non-linear dependencies. Dimensionality reduction via Locally Linear Embedding (LLE) and Uniform Manifold Approximation and Projection (UMAP) further clarifies feature clusters by preserving local and global structures. Clustering models, including Agglomerative Clustering and K-means, highlight feature distributions, showing patterns in rhythmic and harmonic characteristics across songs. Factor Analysis refines the dataset by emphasizing core features that impact popularity, offering insights into latent structures within the data. Factor Analysis condenses the dataset into a limited number of factors and shows the features which contribute most to each factor. However, in Factor Analysis, it focuses on the key features affecting the songs which are most influential in a large dataset. It assists to uncover these X factors that drive the core characteristics behind making a song appealing, thus aligning with the purpose of framework in identifying universals X factor of appeal.

Correlation analysis and mutual information quantified relationships among audio and social features, identifying informative inputs for dimensionality reduction and clustering. LLE preserved local data structures, projecting high-dimensional features into clusters based on similarities, showing that 90% of features are

genre-agnostic. Agglomerative clustering, using LLE findings, hierarchically grouped features into a dendrogram, illustrating associations. UMAP revealed hidden structures, clustering similar genres based on shared attributes, even across diverse sources.

UMAP integrates local and global cluster structures, revealing relationships between features. K-means clustering, known for its scalability and versatility, partitions data into distinct clusters based on proximity to cluster centers. Principal Component Analysis (PCA) on raw feature weights highlights their distribution, helping to understand the variability in audio characteristics across songs.

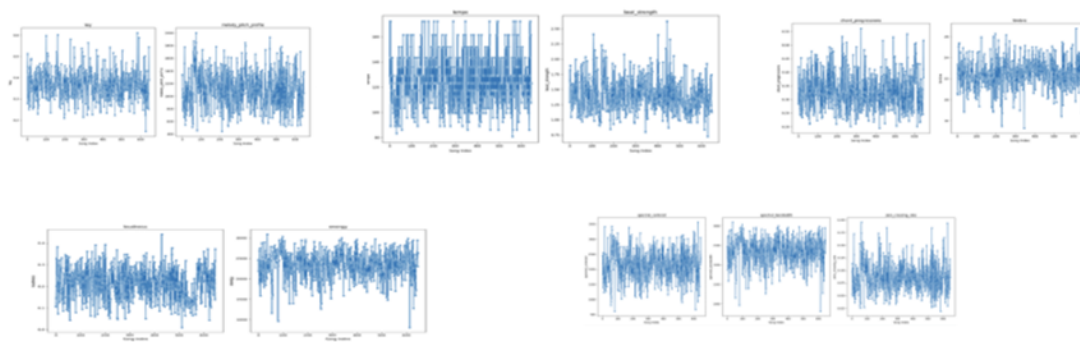
### 3.5 Model Development

Six models were chosen based on their complementary strengths in feature selection, interpretability, and prediction. The varying degrees of complexity inherent in these regression models facilitate an examination of linear as well as non-linear feature interactions, which leads to a better insight on the potential drivers and inhibition factors of a song’s success. The models used are Random Forest, XGBoost, Decision Tree, Support Vector Machine (SVM), Logistic Regression, and the DNN using Attention Mechanism.

### 3.6 Training and Testing

Training and testing procedures utilized the Adam optimizer with a learning rate of 0.001 and categorical cross entropy as the loss function. The model was trained over 100 epochs with a batch size of 32, employing early stopping based on validation loss to prevent overfitting. The training process leveraged both dropout and batch normalization techniques to enhance generalization capability across the diverse audio feature space.

For the Data Splitting Strategy, the dataset was divided with an 80/20 ratio and a fixed random state of 42 for reproducibility. This random sampling approach ensures consistent results across multiple runs while maintaining representative distribution of all popularity classes in both training and test sets.



**Figure 1: Feature Extraction Result**

Evaluation metrics included accuracy, precision, recall, and F1-score for classification performance, alongside mean-squared error (MSE) and R-squared for assessing the model’s predictive performance and consistency. These comprehensive metrics provide multiple perspectives on model effectiveness, with accuracy indicating overall correctness and precision-recall metrics balancing false positives and negatives across the four popularity classes.

### 3.7 Evaluation

The model was evaluated based on accuracy, precision, recall, and F1 score to determine its performance. Each metric provides a different perspective on model performance, with accuracy indicating overall correctness, precision, and recall balancing false positives and negatives, and the F1 score providing a harmonic mean of the latter two. The model’s evaluation phase identifies which approach yields the highest predictive power for song popularity.

## 4 Results and Discussion

This section presents the results from feature extraction and multi-model analysis, along with interpretations of how each model contributed to identifying dominant audio features that correlate with song popularity.

### 4.1 Feature Extraction Results

As shown in Figure 1, feature extraction process identified key audio attributes, including tempo, energy, loudness, and danceability, as primary indicators associated with song popularity. Through the application of the Interquartile Range (IQR) method, outliers were removed to create a more normalized dataset. Min-max scaling was then applied to ensure consistency in the feature range, followed by a comprehensive clustering and dimensionality reduction to highlight relationships among key audio features. The identified features were then analyzed for their predictive value, with energy and loudness emerging as particularly influential in determining song popularity within the sample.

Factor Analysis (FA) in this study aggregates related audio features into composite factors that represent broader musical traits, enhancing the interpretability of feature contributions to song popularity. Key factors emerged: FA1 (Spectral Characteristics), which captures features like energy and spectral centroid, representing

sound sharpness; FA2 (Tonal and Rhythmic Qualities), embodying key, beat strength, and chord progressions that shape a song’s harmonic and rhythmic appeal; and FA3 and FA4 (Rhythm and Melody), which emphasize pacing and melody, crucial for creating memorable tracks. Factors FA5 and FA6, contributing minimal insight, were excluded to refine the model’s focus on impactful traits. This structuring helps clarify how energetic, rhythmically steady, and melodically catchy elements enhance a song’s appeal, particularly in popular genres.

The correlation matrix reveals strong positive relationships, such as between Spectral Bandwidth and Spectral Centroid (both frequency-related) and between Loudness and Energy (indicating that louder songs often feel more intense). Notable negative correlations include Loudness and Instrumentalness, suggesting that instrumental tracks are typically quieter. Mutual Information (MI) analysis complements this by identifying which features most inform the target variable (popularity as measured by YouTube views). Top features by MI include Spectral Centroid, Speechiness, and Loudness, implying that tonal brightness, vocal presence, and energy are key to listener appeal. Danceability, Energy, and Liveness provide moderate predictive value, while features like Instrumentalness and Acousticness contribute minimally.

At the top of the MI rankings are Spectral Centroid, Speechiness, and Loudness, suggesting that these attributes are closely tied to listener engagement; for instance, Spectral Centroid reflects tonal “brightness,” indicating that sharper tonal qualities may enhance appeal. Mid-range features like Danceability, Energy, and Liveness also contribute to predictive power, suggesting that rhythmic qualities, tempo, and a “live” feel positively affect listener preference. Lower-ranked features, including Chord Progressions, Melody Pitch Profile, and Zero Crossing Rate, provide structural or harmonic information but show less impact on listener engagement compared to tonal and rhythmic elements. The lowest MI scores belong to Instrumentalness, Acousticness, and Key, indicating that these features are relatively weak predictors of popularity.

Most of the data points are highly concentrated in clusters, but more so on the right side of the LLE plot. Since this shows that all these information points belong together, patterns being typical across popular clusters probably represent the basic features of songs irrespective of their genre, which is quite aligned with the objectives of finding commonalities that make songs successful.

The clusters that are closer to the center show some overlapping, suggesting that these sets of songs share a core audio feature and may therefore be appealing across the different groups of listeners. Most probably, the central clusters are representative of common traits of popular music, where the songs tend to have broadly relatable features. On the other hand, clusters lying far from the center could depict more unique characteristics, perhaps unique or niche qualities of the audio features. For example, the songs in Cluster 0 could be different from those in Cluster 6 regarding rhythmic structures or tonal qualities that would distinguish them in the dataset. The points that form less dense areas around the edges, particularly those spread out towards the bottom of the plot, indicate songs with more unique feature combinations. These songs may include uncommon tonal or rhythmic qualities, or distinctive audio profiles, setting them apart from the majority. This gradient from dense central clusters to more dispersed points shows that UMAP has successfully highlighted both mainstream patterns and unique deviations within the audio features. The K-Means clustering results, visualized on two principal components, reveal six distinct clusters representing groups of songs with similar audio characteristics. Three clusters (Clusters 2, 4, and 6) form a densely packed central concentration, suggesting that many songs share common audio features, such as balanced tempo, energy, and rhythm, likely resonating with broad listener preferences. These central clusters contain overlapping characteristics, indicating a wide acceptance of these core features. On the periphery, clusters like Cluster 0 and Cluster 8 are more spread out, suggesting these songs have unique or specialized audio traits that may appeal to specific audiences. Cluster variation provides further insights, with tighter clusters (e.g., Cluster 6) indicating homogeneity in audio characteristics, while broader clusters like Cluster 4 display more diversity, potentially reflecting sub-genre differences within the group.

## 4.2 Prototype result

The random forest classifier was able to get 93.98% accuracy in the test dataset and around 83.1% in the validation dataset. This model was able to perform quite well given the PCA features. As you can see in the confusion matrix, most of the prediction revolves around the classes of '0' or the "low view count" class. This is due to the skewness of the distribution of the current trending songs in the Philippines. However, as we can further see, the Random Forest failed to predict classes 3 and 2 for its entirety and was only able to predict 3 out of 5 in the class of '1'.

The logistic regression was only able to get around 61.25% accuracy in the test score, whilst having around 27.70% in the validation dataset. This result is quite low given that the random forest classifier was able to get at least 83.1% of the validation dataset. Furthermore, this logistic regression was able to detect only class 3 data but failed to detect class 2. This class, due to its rareness, gives the research more impact in discerning whether the model works, as desired, irrespective of its accuracy. For the decision tree, it was able to get around 82.46% of accuracy in the test score, while having 59.48% in the validation dataset. While this is quite an acceptable accuracy, only having around 22.98% difference, the highest accuracy of the random forest classifier is still persistent. For the SVM,

it was able to get at most 67.02% in the test dataset having only 32.82% in the validation dataset. This is a very poor performance for the SVM as also seen in the confusion matrix. For the XGBoost, we have 93.97% accuracy in the test dataset and 75.38% in the validation dataset. This is the second-highest performance model we have before the random forest classifier. As you can see, the same works with the random forest have not been able to predict the 3rd class correctly as well as the 2nd class.

The consistent failure of top-performing models (Random Forest and XGBoost) to predict Classes 2 and 3 reflects several characteristics of the Philippine music landscape. These moderate-high view count classes represent songs in transition between niche and mainstream appeal, making them inherently harder to classify based solely on audio features. Mid-tier popularity often depends on cultural factors including lyrics, artist reputation, social media trends-not captured in the model. Additionally, the Philippine streaming market exhibits a "winner-takes-all" pattern where songs either achieve massive popularity or remain in lower tiers, with fewer songs occupying the middle ground. Our clustering analysis supports this, showing that Classes 2-3 contain more diverse audio profiles, suggesting they represent genre-crossing tracks that don't conform to clear audio patterns. This aligns with music industry research indicating that moderate success is often more dependent on non-audio factors than extreme success or failure.

The normal DNN was able to get at most 91.62% accuracy in the test dataset while only having 62.56% in the validation dataset. While this is considered a good performance in terms of our overall tested models, it's still ranked 3 before the SVM in terms of validation performance.

For the attention-based neural network using PCA, we concluded with 90.84% in the test dataset and 53.84% in the validation dataset. For the attention-based neural network using ICA, we concluded with 53.93% accuracy in the test dataset and 31.79% accuracy in the dataset.

## 5 CONCLUSION

The researchers conducted this study to explore the most dominant audio features that characterize the top songs in the case of the Philippines (our dataset contains top Spotify songs between 2018 and 2023). By utilizing data processing, feature extraction, and other machine learning models, the study reveals important audio features that appeal to Filipino listeners. Key features such as tempo, energy, melody, and loudness were highlighted as influential in determining a song's appeal. By employing techniques like Locally Linear Embedding (LLE), Uniform Manifold Approximation and Projection (UMAP), and Agglomerative and Means Clustering, this research was able to uncover patterns and relationships within the dataset, providing a nuanced understanding of feature importance in the Philippine music industry.

The study's implementation of a Deep Neural Network (DNN) prototype with an attention mechanism illustrated how machine learning can be leveraged to prioritize features based on their influence, demonstrating a practical application of feature analysis in music. Beyond mapping regional music, these findings will help artists, producers, and industry personnel to catch up with their

listeners regarding what they like to listen to; they will tell how to be in line with the listeners of those regions.

Considering the findings and limitations of this study, a number of recommendations are made to develop the current framework further and expand its applicability. This framework can be expanded upon in future research to study music preference in other areas of the world, enabling a global, comparative analysis, and allowing some degree of differentiation of the attractiveness of music in different cultural dimensions. This study focused only on chorus sections present in popular songs as they are easier to memorize, and therefore included. However, the use of other sections e.g. verses and bridges can provide a broader perspective on other potential features affecting the popularity of songs.

In addition, broadening the social metrics beyond just YouTube to incorporate streaming data from other platforms and social media may also provide a more comprehensive indication of audience engagement and traction. The DNN model's high accuracy demonstrates the potential for further development using advanced machine learning techniques, such as ensemble methods, to complement feature dimensionality analysis and improve interpretation of feature importance in music popularity prediction. Future research can further advance knowledge of audio feature impacts in the emerging field of music analytics by exploring these areas. This study conducted a comparative evaluation of Convolutional Neural Networks (CNN), Long Short-Term Memory Networks (LSTM), and CNN-LSTM hybrid models for sentiment analysis on online learning-related tweets in the Philippines. The results demonstrate that the CNN-LSTM hybrid model outperforms standalone CNN and LSTM models by effectively capturing both spatial and sequential dependencies in textual data.

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